

University of Dhaka



Assignment - 3

Course Title: Advanced Econometrics-II
Course Code: 408

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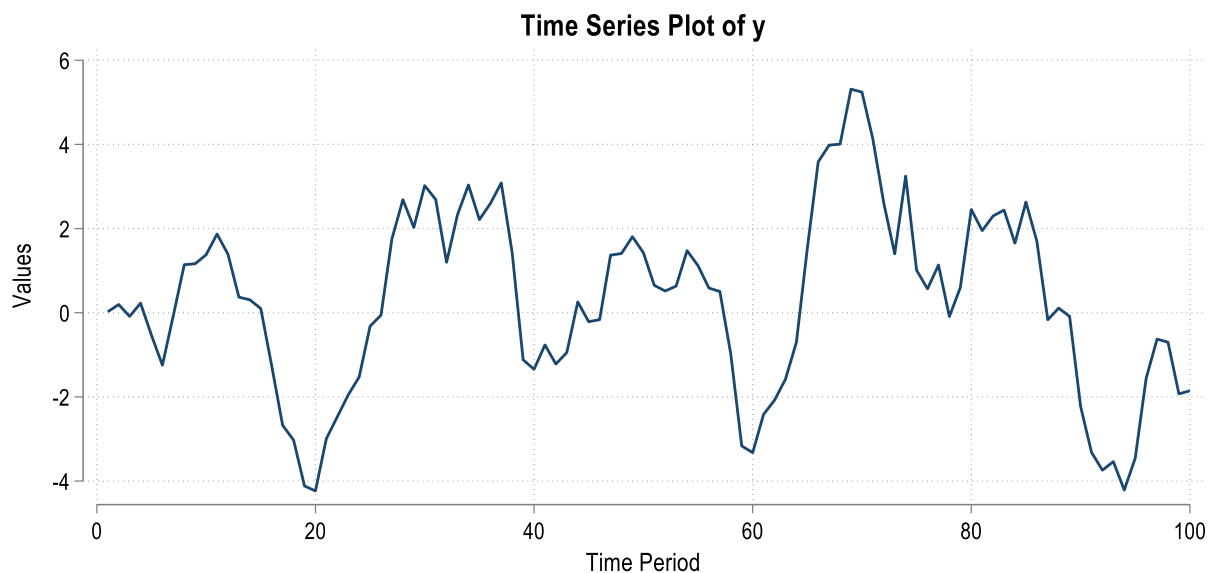
Question:

Attached is a STATA data file with a time series variable "y". First, using the Box-Jenkins methodology, determine which ARMA model this variable follows. After estimating the ARMA model, please forecast 10 steps ahead of "y" with a 95% confidence interval. Extra marks for better visual representation.

Solution:

Following the Box-Jenkins Methodology, we will follow (a) Identification (b) Estimation (c) Diagnostic Test, these 3 procedures to find the best Univariate ARMA model that this variable y follows.

(a) Identification:

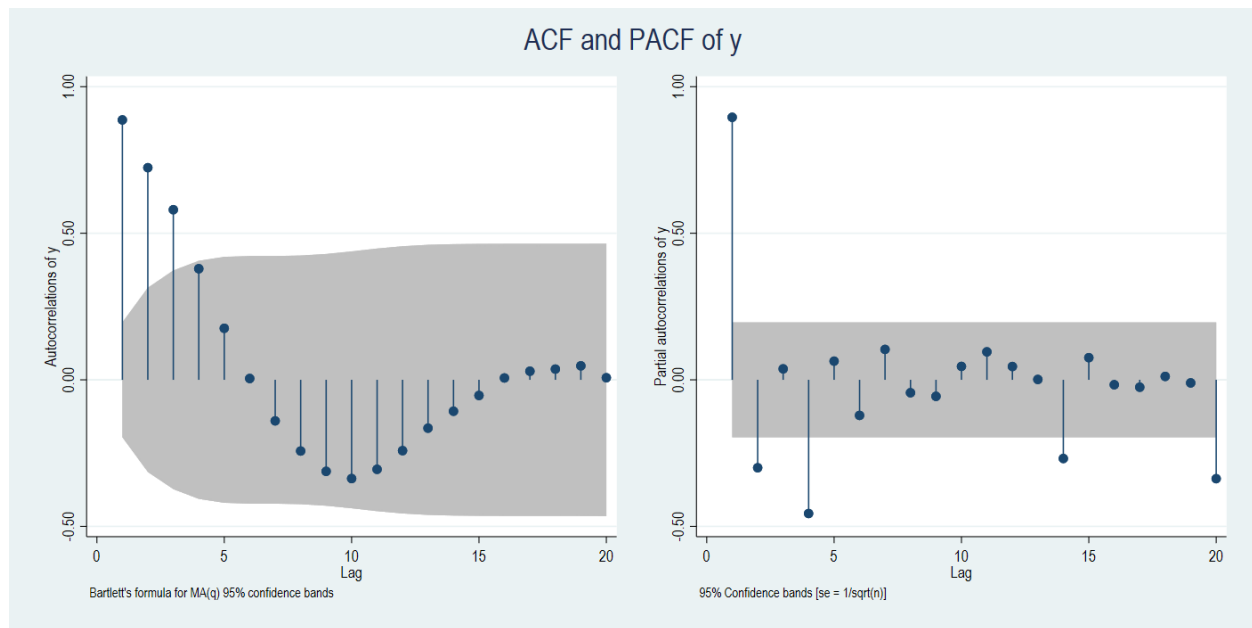


H0: Random walk without drift, $d = 0$

	Test	Dickey Fuller Critical Values		
	Statistics	1%	5%	10%
Z(t)	-3.622	-3.517	-2.894	-2.582

MacKinnon approximate p-value for Z(t) = 0.0054

The Time Series plot of the variable y clearly shows no significant trend. And the Augmented Dickey-Fuller Test also rejects the Null Hypothesis of a Unit Root Process at lag 4. So, we can conclude that variable y is stationary.



Both the Autocorrelation and Partial Auto-correlation Function of y variable show significant spikes till lag 3 and exponential decay in ACF. From Stata's varsoc command I found that the optimal AR lag is 4. So, I assume the presence of AR (4), ARMA (2,3), ARMA (4,1), ARMA (4,2), ARMA (4,3). Now I will estimate all the assumed models and compare their information criteria to get the parsimonious model.

Information Criteria:

<i>Model</i>	<i>N</i>	<i>LL (model)</i>	<i>df</i>	<i>AIC</i>	<i>BIC</i>
AR (4)	100	-124.669	7	261.338	276.969
ARMA (2,3)	100	-124.3935	7	262.7871	281.0233
ARMA (4,1)	100	-124.2639	7	262.5278	280.7639
ARMA (4,2)	100	-123.5619	8	263.1831	284.0245
ARMA (4,3)	100	-123.5017	9	265.0035	288.45

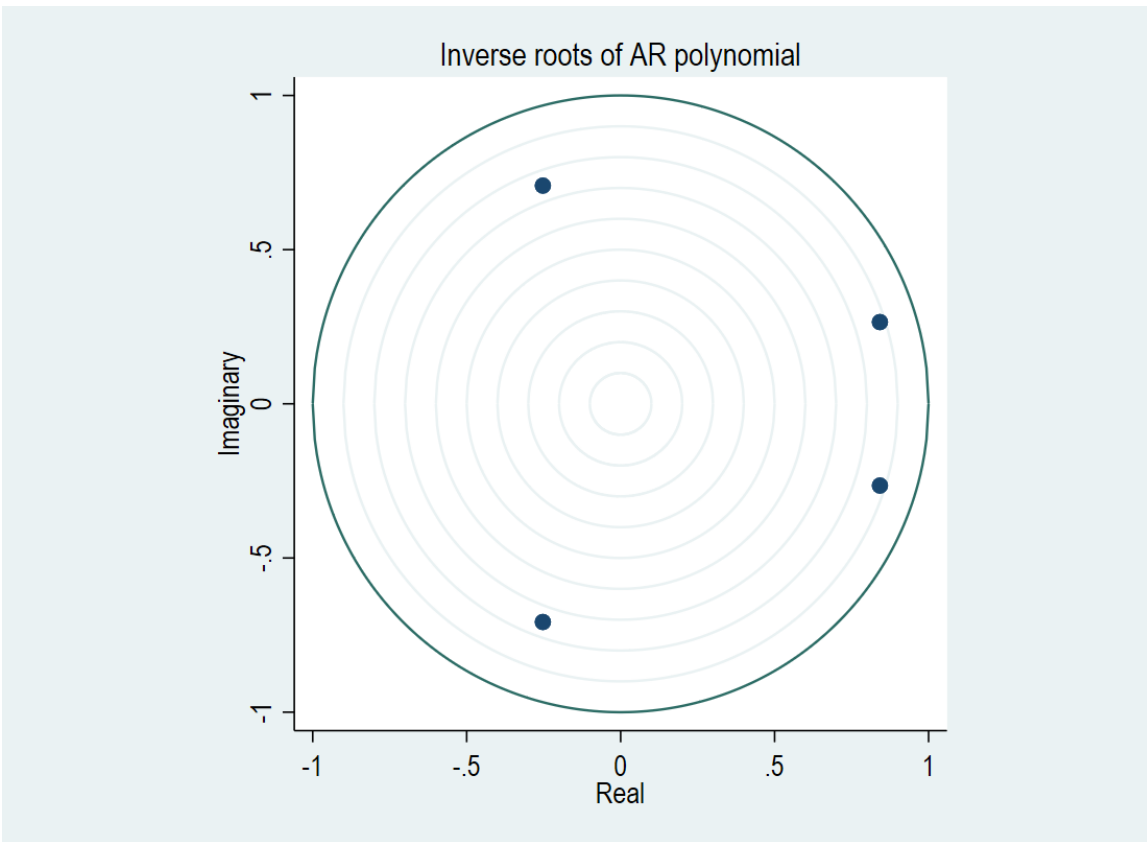
Among those guessed models, AR (4) yields the lowest AIC and BIC so this is my chosen Univariate model for the variable y.

(b) Estimation:

AR (4) / ARMA (4,0) regression:

y	Coef.	St.Err.	t-value	p-value	[95% Conf	Interval]	Sig
Constant	.235	.446	0.53	.598	-.639	1.108	
ar							
L	1.179	.085	13.89	0	1.012	1.345	***
L2	-.493	.122	-4.04	0	-.732	-.254	***
L3	.557	.159	3.50	0	.245	.868	***
L4	-.44	.09	-4.88	0	-.617	-.263	***
Constant	.831	.056	14.97	0	.722	.94	***
Mean dependent var		0.275	SD dependent var			2.162	
Number of obs		100	Chi-square			380.697	
Prob > chi2		.	Akaike crit. (AIC)			261.338	

*** $p < .01$, ** $p < .05$, * $p < .1$



After estimating the AR (4) model we can see all the lag coefficients of AR process are statistically significant at 1% confidence intervals. And the roots diagram shows us that all the roots of AR process are within the unit circle. So, we can say AR (4) or ARMA (4,0) model is acceptable for estimation.

(c) Diagnostic Test:

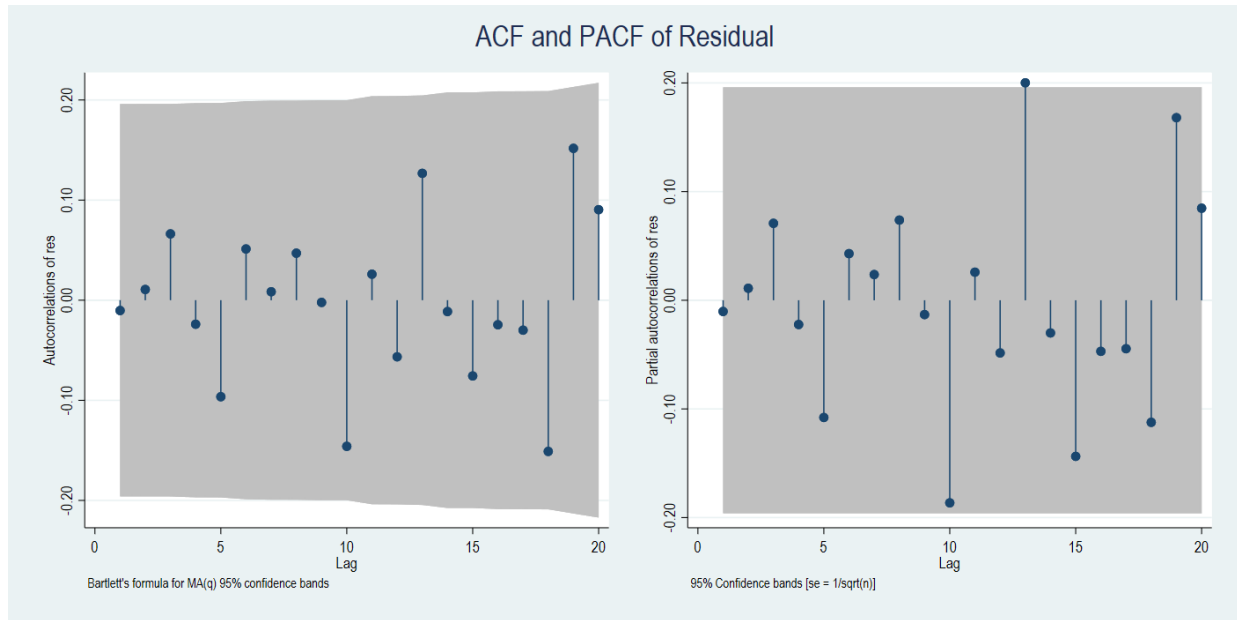
Portmanteau test for H_0 : white noise

Portmanteau (Q) statistics = 33.9823

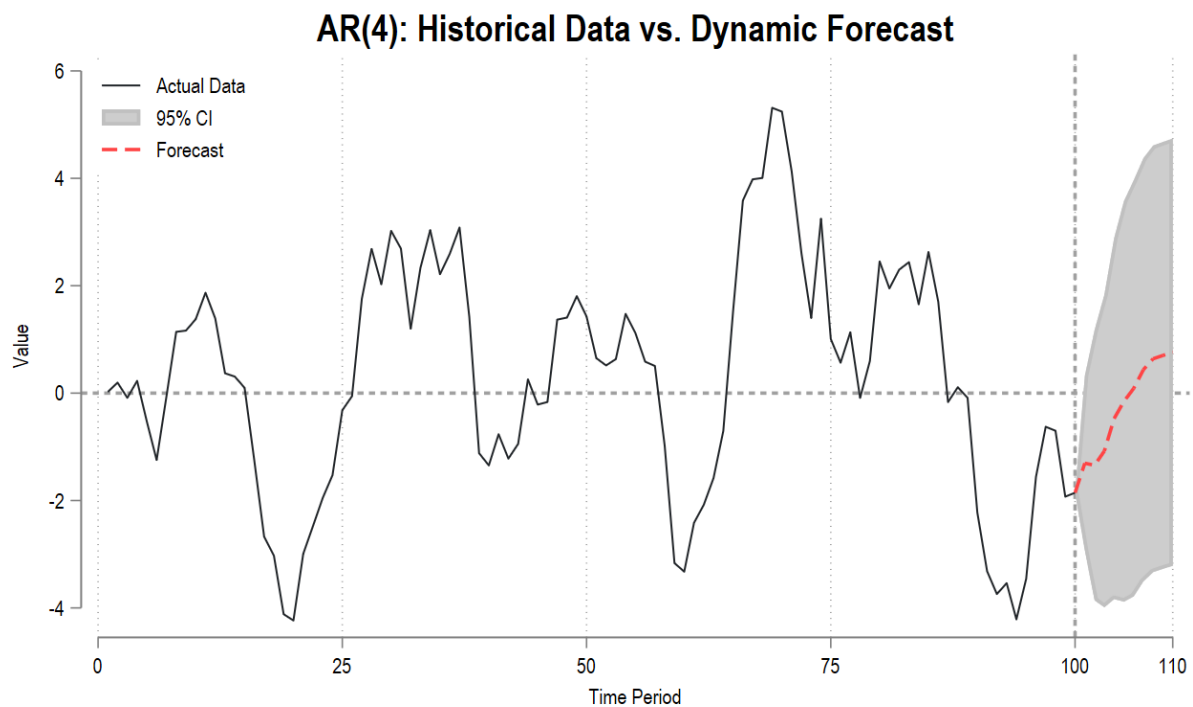
Prob > chi2(40) = 0.7370

The Ljung-Box test or Q test at lag 40 do not reject the Null Hypothesis of the residuals being white noise. The Autocorrelation and Partial Autocorrelation function of the residuals also justify this white noise process for lag 20. This diagnostic test satisfies the residuals being

white noise and according to the steps of Box Jenkins Methodology AR (4) or ARMA (4,0) model is accepted as the univariate process of the variable y.



(d) 10 Steps Ahead Forecast:



```
1  cd "F:\ECON 408\Practice do file"
2
3
4  clear all
5  set seed 1000
6
7
8  use "HOMEWORK 3.dta" , clear
9
10 rename t time
11 tsset time
12
13
14 *                               Identification
15 *****
16
17 *(1) Time Series plot of variable y
18 //-----
19
20
21 set scheme white_tableau
22
23 tsline y, ///
24     lcolor(navy) lwidth(medthick) ///
25     title("{bf:Time Series Plot of y}", size(large)) ///
26     xtitle("Time Period", size(medlarge)) ///
27     ytitle("Values", size(medlarge)) ///
28     ylabel(, labsize(medium)) ///
29     tlabel(, labsize(medium)) ///
30     graphregion(color(white)) xsize(8) ysize(4) ///
31     name(tsplot_hw3, replace)
32
33 graph export tsplot_hw3.png, name(tsplot_hw3) replace
34
35
36 *(2) Stationarity Test
37 //-----
38
39
40 varsoc y , maxlag(12)
41 dfuller y, lags(4)
42
43
44 *(3) ACF and PACF
45 //-----
46
47
48 set scheme s2color
49
50 ac y, name(acf_plot,replace) lags(20)
51 pac y, name(pacf_plot, replace) lags(20)
52 graph combine acf_plot pacf_plot, name(ACF_and_PACF_hw3, replace) ///
53     title("ACF and PACF of y") xsize(8) ysize(4)
54
55 graph export ACF_and_PACF_hw3.png, name(ACF_and_PACF_hw3) replace
56
57 * Assumed models: AR(4), ARMA(2,3), ARMA(4,1), ARMA(4,2), ARMA(4,3)
58
59 qui arima y, ar(1/4)
60 estat ic
61
62 qui arima y, ar(1/2) ma(1/3)
63 estat ic
```

```

64
65 qui arima y, ar(1/4) ma(1)
66 estat ic
67
68 qui arima y, ar(1/4) ma(1/2)
69 estat ic
70
71 qui arima y, ar(1/4) ma(1/3)
72 estat ic
73
74
75 * AR(4) has the lowest AIC and BIC
76
77
78 *
79 ***** Estimation *****
80
81 arima y, ar(1/4)
82 estat ic
83 estat aroots, name(arootshw3,replace)
84 graph export arootshw3.png,name(arootshw3) replace
85
86
87 *
88 ***** Diagnostic Test *****
89
90
91 predict res , residuals
92
93 ac res, name(res_acf, replace) lags(20)
94 pac res , name(res_pacf, replace) lags(20)
95 graph combine res_acf res_pacf , xsize(8) ysize(4) title("ACF and PACF of
96 Residual") ///
97 name(res_acf_pacf_hw3, replace)
98 graph export res_acf_pacf_hw3.png, name(res_acf_pacf_hw3) replace
99
100 wntestq res
101
102
103 *
104 ***** 10 Days Ahead Forecast *****
105
106
107 keep if time <= 100
108 qui arima y, ar(1/4)
109 tsappend, add(10)
110
111 capture drop y_forecast y_mse y_se ci_lower ci_upper
112 predict y_forecast, y dynamic(101)
113 predict y_mse, mse dynamic(101)
114
115 gen y_se = sqrt(y_mse)
116 gen ci_lower = y_forecast - 1.96 * y_se
117 gen ci_upper = y_forecast + 1.96 * y_se
118
119 qui replace y_forecast = y if time == 100
120 qui replace ci_lower = y if time == 100
121 qui replace ci_upper = y if time == 100
122
123 list time y_forecast ci_lower ci_upper if time >=99 & time <= 103
124
125

```



```
126 set scheme white_tableau
127
128 twoway (rarea ci_lower ci_upper time if time >= 100, color("gs12") lwidth(
medthick)) ///
129       (line y time if time <= 100 & time >= 1, lcolor("33 37 41") lwidth(medthin
)) ///
130       (line y_forecast time if time >= 100, lcolor("255 68 69") lpattern(dash)
lwidth(medthick)), ///
131       xline(100, lcolor(gs10) lwidth(medthick) lpattern(shortdash)) ///
132       yline(0, lcolor(gs10) lwidth(medthick) lpattern(shortdash)) ///
133       graphregion(color(white)) ///
134       title("{bf:AR(4): Historical Data vs. Dynamic Forecast}") ///
135       xtitle("Time Period") ///
136       ytitle("Value") ///
137       legend(order(2 "Actual Data" 1 "95% CI" 3 "Forecast") pos(11) ring(0) ///
cols(1)) ///
138       ylabel(, format(%9.0f) nogrid) ///
139       xlabel(0(25)100 110, labsize(small)) ///
140       xsize(6.5) ysize(4) name(forecast_plot,replace)
141
142
143 graph export forecast_plot.png, name(forecast_plot) replace
144
145
146
```